BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

BOTANY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A (Multiple Choice Questions (MCQ's))

Note: Q.1 Choose the correct answer for each from the given options:

Q1. In double fertilization, one sperm fuses with _____ and other with _____.

(a) egg- synergids
(b) egg- antipodal cell
(c) egg – two polar nuclei
(d) egg- one polar nucleus

Q2. In double stranded DNA molecule, the amount of Thymine is equal to the amount of _____.

(a) Uracil
(b) Guanine
(c) Adenine
(d) Cytosine

Q3. The production of seeds without fertilization called _____.

(a) Parthenocarpy
(b) Apomixis
(c) Parthenogenesis
(d) Sporulation

Q4. All are examples of sexual reproduction in plants except.

(a) Cutting
(b) Tissue culture
(c) Sproulation
(d) Cloning

Q5. The plants which grow partly or completely sub-merged in fresh water are termed _____.

(a) [illegible in source scan]
(b) [illegible in source scan]
(c) [illegible in source scan]
(d) [illegible in source scan]

Q6. All are examples of sexual reproduction in plants except.

(a) Isogamy
(b) anis gamy
(c) Hydrophytes
(d) Biosphere

Q7. Group of different species living together in same environment termed as _____.

(a) Population
(b) Community
(c) Ecosystem
(d) Biosphere

Q8. At what stage of mitosis, the chromosomes arrange them selves at the equatorial plane?

(a) Telophase
(b) Metaphase
(c) Anaphase
(d) Prophase

Q9. The capacity of living system to lose water is called _____

(a) Osmosis
(b) Plasmolysis
(c) Water potential
(d) osmotic pressure

Q10. The movemnt of lant organs in the response to touch stimulus is called _____.

(a) Geotropism
(b) Thigmotropism
(c) Chemotorpism
(d) Hydrotorpism

Q11. Increase in diameter of stem and root is called _____

(a) Pimary growth
(b) Secondary growth
(c) Teriary growth
(d) growth correlation.

Q12. All are the climatic factors of ecosystem except _____

(a) Light
(b) Temperature
(c) Altitude
(d) Water

Q13. In _____ zone of ocean light penetrates and producers can prepare their food by photosynthesis.

(a) Abyssal
(b) Benthic
(c) Euphotic
(d) Peltgic

Q14. Mitosis provides a means for all the following except.

(a) Healing of wound
(b) Development
(c) Evolution of species
(d) Growth

Q15. About _____ genes are present on human X chromosome.

(a) 200
(b) 400
(c) 600
(d) 1000

Q16. During pollination, pollens travel from anther to the _____ of flower.

(a) Style
(b) Stigma
(c) Filament
(d) Ovary

Q17. halophytes have following characteristics except.

(a) Growing in salt marshes
(b) Found in fresh water
(c) Having salt gland on leaves
(d) Trap water vapours from air

Q18. Where proteins are made in the cell?

(a) Nucleus
(b) Ribosomes
(c) Mitochondira
(d) Golgi bodies

Q19. The division of cytoplasms is called _____

(a) Cuytokinins
(b) Karyokinesis
(c) Synapsis
(d) cytokinesis

Q20. After fertiliziation, ovule develops into _____

(a) Zygote
(b) Flower
(c) Fruit
(d) Seed

Q21. The semiforest grasslands of tropical regions are called _____

(a) Desert
(b) undra
(c) Savannah
(d) Bread-basket

Q22. Ponds, lakes and rivers are the reservoirs of _____

(a) Sea water
(b) Fresh water
(c) Both (a) and (b)
(d) Brakish water

Section-B

> Note: Answer any Three of the following questions. Each question carries 02 marks.

Q2. How HALOPHYTES absorb water from the having lower water potential than the cell-sap?

Q3. Do the cells beneath formative-phase differ from each other?

Q4. What in Mutation?

Q5. How HETERO-CHROMATIN differs form the EU-CHROMATIN?

Q6. How bisexual flowers are prevented from self-pollination and to favours cross-pollination?

Q7. Why TRANSCRIPTION and TRANSLATION are so named?

Section-C (Descriptive Answer)

Q8. How abnormal Eukarytic cell differs form normal Eukarytoic cell?

Q9. What are Heliophytes and Sciophytes?

Q10. What is DIHYBRID cross? Describe the law of INHERTIANCE which explains the inheritance of dihybrid cross?